

Let $X = \{1, \dots, n\}$. For each $\alpha \in \{1, \dots, n\}$, let $\hat{\alpha} : C(X) \rightarrow \mathbb{C}$ be given by $\hat{\alpha}(f) := f(\alpha)$. Then $\Sigma_{C(X)} = \{\hat{1}, \hat{2}, \dots, \hat{n}\}$, where $\Sigma_{C(X)}$ is the set of all nonzero algebra homomorphisms from $C(X)$ to $\mathbb{C}$ (i.e., the characters of $C(X)$).

First, note that $\{\delta_1, \dots, \delta_n\}$ is a basis for $C(X)$, where each δ_i is defined by:

$$\delta_i(j) = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}.$$

Clearly $\{\hat{1}, \hat{2}, \dots, \hat{n}\} \subseteq \Sigma_{C(X)}$, as for any $\hat{\alpha} \in \{\hat{1}, \hat{2}, \dots, \hat{n}\}$ we have:

$$\begin{aligned} \hat{\alpha}(f + g) &= (f + g)(\alpha) = f(\alpha) + g(\alpha), \\ \hat{\alpha}(fg) &= (fg)(\alpha) = f(\alpha)g(\alpha), \\ \hat{\alpha}(\lambda f) &= (\lambda f)(\alpha) = \lambda f(\alpha). \end{aligned}$$

Moreover, each $\hat{\alpha}$ is nonzero, as $\hat{\alpha}(\delta_\alpha) = \delta_\alpha(\alpha) = 1$.

Claim: for any $h \in \Sigma_{C(X)}$, there is exactly one $\delta_i \in \{\delta_1, \dots, \delta_n\}$ such that $h(\delta_i) = 1$, and $h(\delta_j) = 0$ for every $j \neq i$. Since $\delta_i^2 = \delta_i$, we have $h(\delta_i)^2 = h(\delta_i^2) = h(\delta_i)$. Hence $h(\delta_i) = 1$ or 0. If it were the case that $h(\delta_i) = 0$ for all $i = 1, \dots, n$, then h would be the zero functional which contradicts it being an element of $\Sigma_{C(X)}$. Now consider $\sum_{\alpha=1}^n \delta_\alpha$. It is also true that $(\sum_{\alpha=1}^n \delta_\alpha)^2 = \sum_{\alpha=1}^n \delta_\alpha$, hence:

$$\left(\sum_{\alpha=1}^n h(\delta_\alpha) \right)^2 = \sum_{\alpha=1}^n h(\delta_\alpha).$$

It must be the case that $\sum_{\alpha=1}^n h(\delta_\alpha) = 1$. Since $h(\delta_\alpha)$ can be either 1 or 0, this is only true when $h(\delta_\alpha) = 1$ for exactly one $\delta_\alpha \in \{\delta_1, \dots, \delta_n\}$.

With this, let $h \in \Sigma_{C(X)}$. Suppose i is the unique index such that $h(\delta_i) = 1$. Then for any $f \in C(X)$:

$$\begin{aligned} h(f) &= h\left(\sum_{\alpha=1}^n f(\alpha)\delta_\alpha\right) \\ &= \sum_{\alpha=1}^n f(\alpha)h(\delta_\alpha) \\ &= f(i) \\ &= \hat{i}(f). \end{aligned}$$

Thus $h \in \{\hat{1}, \hat{2}, \dots, \hat{n}\}$, establishing $\Sigma_{C(X)} = \{\hat{1}, \hat{2}, \dots, \hat{n}\}$.